CHAPTER 29

The Science and Design of Assessment in Engineering Education

James W. Pellegrino, Louis V. DiBello, and Sean P. Brophy

Chapter Overview and Goals

In 2001, a report was issued by the National Research Council (NRC) entitled "*Knowing What Students Know: The Science and Design of Educational Assessment*" (Pellegrino, Chudowsky, & Glaser, 2001). The goal was to evaluate the state of research and theory on educational assessment and establish the scientific foundations for their design and use. As argued in that volume, many of the debates that surround educational assessment emanate from a failure to understand its fundamental nature, including the ways in which theories and models (1) of learning and knowing and (2) of measurement and statistical inference interact with and influence processes of assessment design, use, and interpretation. In this chapter we review some of the key issues regarding educational assessment raised in that report as well as examples from science, technology, engineering, and mathematics (STEM) education and educational research contexts. Our goal in explicating current understanding of the science and design of educational assessment and its applications to STEM education is to provide background knowledge that supports the effective design and use of assessment in engineering education and sharpens the focus of engineering education R&D.

In the first section we briefly introduce some key ideas critical to understanding educational assessment. This includes consideration of formal and informal uses of assessment and some conceptual issues associated with assessment design, interpretation, and use. In the second section we discuss three related conceptual frameworks about assessment that should be considered by anyone using assessment for instructional or research purposes. These include (1) assessment as a process of reasoning from evidence, (2) the use of an evidence-centered design process to develop and interpret assessments, and (3) centrality of the concept of validity in the design, use, and interpretation of any assessment. In the third section we turn to a discussion of concepts of measurement as applied to assessment and the role of statistical inference. This includes the assumptions underlying different types of psychometric models used to estimate

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

student proficiency. The fourth section then presents applications of key ideas from the preceding sections in the form of illustrative examples of assessments used in engineering education research. In the final section we close by briefly considering the significance of a careful and thoughtful approach to assessment design, use, and interpretation in engineering education research.

Educational Assessment in Context

Assessment Purposes and Contexts

From teachers' classroom quizzes, mid-term or final exams, to nationally and internationally administered standardized tests, assessments of students' knowledge and skills have become a ubiquitous part of the educational landscape. Assessments of school learning provide information to help educators, administrators, policymakers, students, parents, and researchers judge the state of student learning and make decisions about implications and actions. The specific purposes for which an assessment will be used are an important consideration in all phases of its design. For example, assessments used by instructors in classrooms to assist or monitor learning typically need to provide more detailed information than assessments whose results will be used by policymakers or accrediting agencies. One of the central points of the *Knowing What Students Know* report was that assessments are developed for specific purposes, and the nature of their design is very much constrained by their intended interpretive use.

ASSESSMENT TO ASSIST LEARNING

In the classroom context, instructors use various forms of assessment to inform day-to-day and month-to-month decisions about next steps for instruction, to give students feedback about their progress, and to motivate students. One familiar type of classroom assessment is a teacher-made quiz, but assessment also includes more informal methods for determining how students are progressing in their learning, such as classroom projects, feedback from computer-assisted instruction, classroom observation, written work, homework, and conversations with and among students – all interpreted by the teacher in light of additional information about the students, the schooling context, and the content being studied.

These situations are referred to as *assessments to assist learning*, or the *formative use of assessment*. These assessments provide specific information about students' strengths and difficulties with learning. For example, statistics teachers need to know more than the fact that a student does not understand probability; they need to know the details of this misunderstanding, such as the student's tendency to confuse conditional and compound probability. Teachers can use information from these types of assessments to adapt their instruction to meet students' needs, which may be difficult to anticipate and are likely to vary from one student to another. Students can use this information to determine which skills and knowledge they need to study further and what adjustments in their thinking they need to make.

ASSESSMENT OF INDIVIDUAL ACHIEVEMENT

Another type of assessment used to make decisions about individuals is that conducted to help determine whether a student has attained a certain level of competency after completing a particular phase of education, whether it be a two-week curricular unit, a semester long course, or twelve years of schooling. This is referred to as *assessment of individual achievement*, or the *summative use of assessment*. Some of the most familiar forms of summative assessment are those used by classroom instructors, such as end-of-unit or end-of-course tests, which often are used to assign letter grades when a course is finished. Large-scale assessments – which are administered at the direction of users external to the classroom – also provide information about the attainment of individual students, as well as comparative information about how one individual performs relative to others. Because large-scale

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

assessments are typically given only once a year and involve a time lag between testing and availability of results, the results seldom provide information that can be used to help teachers or students make day-to-day or month-to-month decisions about teaching and learning.

ASSESSMENT TO EVALUATE PROGRAMS

Another common purpose of assessment is to help administrators, policymakers, or researchers formulate judgments about the quality and effectiveness of educational programs and institutions. Instructional evaluation can be considered formative in nature when used to improve the effectiveness of instruction. Summative uses of assessment for evaluation are incorporated increasingly in making high-stakes decisions not only about individuals but also about programs and institutions. For instance, public reporting of state assessment results by school and district can influence the judgments of parents and taxpayers about the quality and efficacy of their schools and affect decisions about resource allocations. Just as with individuals, the quality of the measure is of critical importance in the validity of these decisions.

Further Considerations of Purposes, Levels, and Timescales

As noted previously, assessment occurs in multiple contexts, has a variety of formal and informal uses, and is conducted to meet different purposes. The purpose of an assessment determines priorities, and the context of use imposes constraints on the design. Thus it is essential to recognize that one type of assessment does not fit all purposes or contexts of use. In general, the more purposes a single assessment aims to serve, the more each purpose will be compromised and the overall product will represent a suboptimal design for each intended use. A persistent mistake is to assume that an assessment is appropriate and interpretable for a particular context of use without determining if there is evidence regarding the validity of such assumptions (see later discussions of validity) within that context. The one-size-fits-all fallacy is especially frequent and problematic since it produces inappropriate choices of assessments for instructional or research purposes that in turn can lead to invalid conclusions regarding persons, programs, and/or institutions.

Although assessments are currently used for many purposes in the educational system, a premise of the *Knowing What Students Know* report is that their effectiveness and utility must ultimately be judged by the extent to which they promote student learning. The aim of assessment should be "*to educate and improve* student performance, not merely to *audit* it" (Wiggins, 1998, p. 7). Because assessments are developed for specific purposes, the nature of their design is very much constrained by their intended use. Although it may seem reasonable to dichotomize between *internal* classroom assessments, administered by instructors, and *external* tests, administered by districts, states, or nations or other agencies, such a dichotomy is an oversimplification of a continuum that reflects the proximity of an assessment to the enactment of specific instructional and learning activities. Ruiz-Primo, Shavelson, Hamilton, and Klein (2002) defined five discrete points on a continuum of assessment distance: *immediate* (e.g., observations or artifacts from the enactment of a specific instructional activity), *close* (e.g., embedded assessments and semiformal quizzes of learning from one or more activities), *proximal* (e.g., formal classroom exams of learning from a specific curriculum), *distal* (e.g., criterion-referenced achievement tests such as required by the federal No Child Left Behind legislation), and *remote* (broader outcomes measured over time, including norm-referenced achievement tests and some national and international achievement measures). Different assessments should be understood as different points on this continuum if they are to be aligned effectively with each other and with curriculum and instruction. In essence, an assessment is a test of transfer and it can be near or far transfer depending

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

on where the assessment falls along the continuum noted earlier.

The level at which an assessment is intended to function, which involves varying distance in "space and time" from the enactment of instruction and learning, has implications for how and how well it can fulfill various functions of assessment, be they formative, summative, or program evaluation (NRC, 2003). As argued elsewhere (Hickey & Pellegrino, 2005; Pellegrino & Hickey, 2006), it is also the case that the different levels and functions of assessment can have varying degrees of match with theoretical stances about the nature of knowing and learning (see also Chapter 2 Newstetter & Svinicki in this volume).

Although assessments used in various contexts and for differing purposes and at different timescales often look quite different, they share certain common principles. One such principle is that assessment is always a process of reasoning from evidence. By its very nature, moreover, assessment is imprecise to some degree. Assessment results are only estimates of what a person knows and can do. We elaborate on both of these issues in the following two sections.

Conceptual Frameworks

Assessment as a Process of Evidentiary Reasoning: The Assessment Triangle

Educators assess students to learn about what they know and can do, but assessments do not offer a direct pipeline into a student's mind. Assessing educational outcomes is not as straightforward as measuring height or weight; the attributes to be measured are mental representations and processes that are not outwardly visible. Thus, an assessment is a tool designed to observe students' behavior and produce data that can be used to draw reasonable inferences about what students know. Deciding what to assess and how to do so is not as simple as it might appear.

The process of collecting evidence to support inferences about what students know represents a chain of reasoning from evidence about student learning that characterizes all assessments, from classroom quizzes and standardized achievement tests, to computerized tutoring programs, to the conversation a student has with his or her teacher as they work through a math problem or discuss the meaning of a text. People reason from evidence every day about any number of decisions, small and large. When leaving the house in the morning, for example, one does not know with certainty that it is going to rain, but may reasonably decide to take an umbrella on the basis of such evidence as the morning weather report and the threatening clouds in the sky.

The first question in the assessment reasoning process is "evidence about what?" *Data* become *evidence* in an analytic problem only when one has established their relevance to a conjecture being considered (Schum, 1987, p. 16). Data do not provide their own meaning; their value as evidence can arise only through some interpretational framework. What a person perceives visually, for example, depends not only on the data she receives as photons of light striking her retinas, but also on what she thinks she might see. In the present context, educational assessments provide data such as written essays, marks on answer sheets, presentations of projects, or students' explanations of their problem solutions. These data become evidence only with respect to conjectures about how students acquire knowledge and skill.

In the *Knowing What Students Know* report the process of reasoning from evidence was portrayed as a triad of three interconnected elements: the *assessment triangle*. The vertices of the assessment triangle (see Figure 29.1) represent the three key elements underlying any assessment: a model of student *cognition* and learning in the domain of the assessment; a set of assumptions and principles about the kinds of *observations* that will provide evidence of students' competencies; and an *interpretation* process for making sense of the evidence in light of the assessment purpose and student understanding. These three elements may be explicit

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

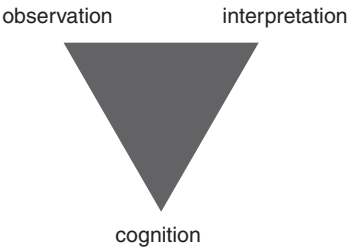


Figure 29.1. The assessment triangle.

or implicit, but an assessment cannot be designed and implemented, or evaluated, without consideration of each. The three are represented as vertices of a triangle because each is connected to and dependent on the other two. A major tenet of the *Knowing What Students Know* report is that for an assessment to be effective and valid, the three elements must be in synchrony. The assessment triangle provides a useful framework for analyzing the underpinnings of current assessments to determine how well they accomplish the goals we have in mind, as well as for designing future assessments and establishing validity (e.g., see Marion & Pellegrino, 2006).

The *cognition* corner of the triangle refers to theory, data, and a set of assumptions about how students represent knowledge and develop competence in a subject matter domain (e.g., fractions, Newton's laws, thermodynamics). In any particular assessment application, a theory of learning in the domain is needed to identify the set of knowledge and skills that is important to measure for the intended context of use, whether that be to characterize the competencies students have acquired at some point in time to make a summative judgment or to make formative judgments to guide subsequent instruction so as to maximize learning. A central premise is that the cognitive theory should represent the most scientifically credible understanding of typical ways in which learners represent knowledge and develop expertise in a domain.

Every assessment is also based on a set of assumptions and principles about the kinds of tasks or situations that will prompt students to say, do, or create something that demonstrates important knowledge and skills. The tasks to which students are asked to respond on an assessment are not arbitrary. They must be carefully designed to provide evidence that is linked to the cognitive model of learning and to support the kinds of inferences and decisions that will be made on the basis of the assessment results. The *observation* vertex of the assessment triangle represents a description or set of specifications for assessment tasks that will elicit illuminating responses from students. In assessment, one has the opportunity to structure some small corner of the world to make observations. The assessment designer can use this capability to maximize the value of the data collected, as seen through the lens of the underlying assumptions about how students learn in the domain.

Every assessment is also based on certain assumptions and models for interpreting the evidence collected from observations. The *interpretation* vertex of the triangle encompasses all the methods and tools used to reason from fallible observations. It expresses how the observations derived from a set of assessment tasks constitute evidence about the knowledge and skills being assessed. In the context of large-scale assessment, the interpretation method is usually a statistical model, which is a characterization or summarization of patterns one would expect to see in the data given varying levels of student competency (see the section "Issues of Measurement: Interpretation and Statistical Inference"). In the context of classroom assessment, the interpretation is often made less formally by the teacher, and is often based on an intuitive or qualitative model rather than a formal statistical one. Even informally teachers make coordinated judgments about what aspects of students' understanding and learning are relevant, how a student has performed one or more tasks, and what the performances mean about the student's knowledge and understanding.

A crucial point is that each of the three elements of the assessment triangle not only must make sense on its own, but also must connect to each of the other two elements

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

in a meaningful way to lead to an effective assessment and sound inferences. Thus, to have an effective assessment, all three vertices of the triangle must work together in synchrony. Central to this entire process, however, are theories and data on how students learn and what students know as they develop competence for important aspects of the curriculum.

Assessment Development: Evidence-Centered Design

Although it is especially useful to conceptualize assessment as a process of reasoning from evidence, the design of an actual assessment is a challenging endeavor that needs to be guided by theory and research about cognition as well as practical prescriptions regarding the processes that lead to a productive and potentially valid assessment for a particular context of use. As in any design activity, scientific knowledge provides direction and constrains the set of possibilities, but it does not prescribe the exact nature of the design, nor does it preclude ingenuity to achieve a final product. Design is always a complex process that applies theory and research to achieve near-optimal solutions under a series of multiple constraints, some of which are outside the realm of science. In the case of educational assessment, the design is influenced in important ways by variables such as its purpose (e.g., to assist learning, to measure individual attainment, or to evaluate a program), the context in which it will be used (classroom or large-scale), and practical constraints (e.g., resources and time).

The tendency in assessment design is to work from a somewhat "loose" description of what it is that students are supposed to know and be able to do (e.g., standards or a curriculum framework) to the development of tasks or problems for them to answer. Given the complexities of the assessment design process, it is unlikely that such a loose process can lead to generation of a quality assessment without a great deal of artistry, luck, and trial and error. As a consequence, many assessments are insufficient on a

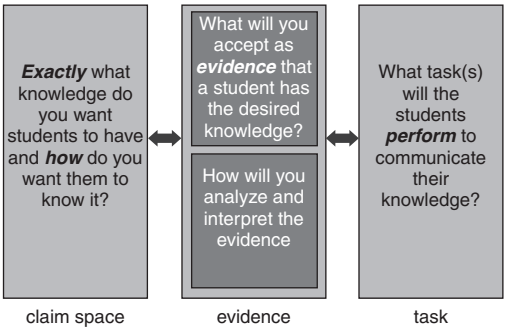


Figure 29.2. Simplified representation of three critical components of the evidence-centered design process and their reciprocal relationships.

number of dimensions including representation of the cognitive constructs and content to be covered and uncertainty about the scope of the inferences that can be drawn from task performance.

Recognizing that assessment is an evidentiary reasoning process, it has proven useful to be more systematic in framing the process of assessment design as an evidence-centered design process (e.g., Mislevy & Haertel, 2006; Mislevy & Riconscente, 2006). Figure 29.2 suffices to capture three essential components of the overall process. As shown in the figure, the process starts by defining as precisely as possible the claims that one wants to be able to make about student knowledge and the ways in which students are supposed to know and understand some particular aspect of a content domain. Examples might include aspects of algebraic thinking, ratio and proportion, force and motion, heat and temperature, and so forth. The most critical aspect of defining the claims one wants to make for purposes of assessment is to be as precise as possible about the elements that matter and express these in the form of verbs of cognition that are much more precise and less vague than high-level cognitive superordinate verbs such as *know* and *understand*. Example verbs might include compare, describe, analyze, compute, elaborate, explain, predict, justify, and so forth. Guiding this process of specifying the claims is theory and research on the nature of domain-specific knowing and learning

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

(see Chapter 2 by Newstetter & Svinicki in this volume).

Although the claims one wishes to make or verify are about the student, they are linked to the forms of evidence that would provide support for those claims – the warrants in support of each claim. The evidence statements associated with given sets of claims capture the features of work products or performances that would give substance to the claims. This includes which features need to be present and how they are weighted in any evidentiary scheme – that is, what matters most and what matters least or not at all. For example, if the evidence in support of a claim about a student's knowledge of the laws of motion is that the student can analyze a physical situation in terms of the forces acting on all the bodies, then the evidence might be a free body diagram that is drawn with all the forces labeled including their magnitudes and directions.

The precision that comes from elaborating the claims and evidence statements associated with a domain of knowledge and skill pays off when one turns to the design of tasks or situations that can provide the requisite evidence. In essence, tasks are not designed or selected until it is clear what forms of evidence are needed to support the range of claims associated with a given assessment situation. The tasks need to provide all the necessary evidence and they should allow students to "show what they know" in a way that is as unambiguous as possible with respect to what the task performance implies about student knowledge and skill – that is, the inferences about student cognition that are permissible and sustainable from a given set of assessment tasks or items.

Assessment Validity: Argumentation and Evidence

The joint AERA/APA/NCME Standards (1999) frame validity largely in terms of "the concept or characteristic that a test is designed to measure" (1999, p. 5). In Messick's construct-centered view of validity, the theoretical construct the test score is purported to represent and score use are the foundations for interpreting the validity of any given test (Messick, 1994). For Messick, validity is "an integrated evaluative judgment of the degree to which empirical evidence and theoretical rationales support the adequacy and appropriateness of inferences and actions based on test scores" (1989, p. 13). Moving beyond a construct-centered view of validity, a more contemporary perspective calls for an interpretive validity argument, an argument that "specifies the proposed interpretations and uses of test results by laying out the network of inferences and assumptions leading from the observed performances to the conclusions and decisions based on the performances" (Kane, 2006, p. 23).

Kane (2006) and others (Haertel & Lorie, 2004; Mislevy, Steinberg, & Almond, 2003), distinguish between the interpretive argument, that is, the propositions that underpin test score interpretation, and the evidence and arguments that provide the necessary warrants for the propositions or claims of the interpretive argument. This view has been elaborated by Mislevy et al (2003) by asserting that validity evidence can be represented more formally in terms of a structured argument, an approach initially introduced and outlined in the context of legal reasoning by Stephen Toulmin (2003). In Toulmin's view a good argument is one that provides strong evidentiary support for a claim that, in turn, permits it to withstand criticism. Toulmin's practical approach begins by explicating the various "claims of interest" and then provides justification for those claims by identifying the evidence (data and/or expert opinion) to support those claims. The approach also calls for supplying a "warrant" that interprets the data and explicates how the data support the claims of interest. Toulmin's work, historically, focused on the formal role of argumentation and was used to evaluate the rational basis of arguments presented typically in the courtroom. Appropriating this approach, contemporary educational measurement theorists have more recently framed test validity as a reasoned

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

argument backed by evidence (e.g., Kane, 2006). The particular forms of evidence are thus associated with the claims that one wishes to make about what a given assessment is and does and how its scores are to be interpreted.

Issues of Measurement: Interpretation and Statistical Inference

Measurement's Role in Evidentiary Reasoning

We have conceptualized assessments as evidentiary reasoning systems specifically constructed for making inferences about states of student knowledge (to be formally represented by latent variables) from observable evidence. The goal of an assessment can be construed in general terms as making particular inferences about examinee knowledge and thinking based on examinee performance on specific tasks. By design, the desired inferences can be expressed in the form of claims about what examinees know and can do under given circumstances. As part of assessment design, once the claims have been articulated explicitly, we can define a formal probability space called the student model as a unidimensional or multidimensional space of proficiency variables in such a way that each student can be thought of as possessing one set of values of the proficiency variables. Inferences about the values of student model variables can be interpreted as claims about students' knowledge and understanding.

Operationally, an examinee responds to a number of predesigned tasks. The examinee's performance results in some "work product" produced by the examinee and consisting of the observable trace left behind by the examinee, which hopefully reflects her thinking and understanding. A work product for a single task could be as simple as a multiple choice option selected – the examinee selected response option c on question 13 – or as complex and extended as a full-blown essay or a set of lab notes for a scientific experiment including drawings, equations, and text. In either case the work product is whatever is produced by the examinee and left behind to be observed as her assessment performance.

An examinee's work product is then "scored" in some way according to a local evidence model. In the case of a multiple-choice option, the question response option c might be scored simply as right or wrong according to an answer key. For a concept inventory, a multiple-choice option may be linked to a specific misconception or common student error. In that case the "score" for the question response may be the misconception that was linked to the examinee's answer choice. For example, the examinee chose option c to question 13 and option c is linked to misconception 5, so we score the response to this question as misconception 5.

For questions with extended responses the examinee's work product for the responses may present multiple features that are observable and that are related to the knowledge intended to be measured by the item. In such a case, the "score" for the question response may consist of multiple feature scores of the work product, one such feature score for each of the observable features that has been identified for relevance to the construct measured and flagged for scoring. Alternatively, a more "holistic" rubric with multiple scoring categories (e.g., scores of 1–5) may be applied to the product (e.g., an essay or explanation) and a single score assigned to the work product.

The heart of the assessment system is the evidentiary reasoning step whereby we make inferences about examinee proficiencies (which can then be translated into claims about the examinee) based on evidence represented by the question scores as described earlier. The fundamental Bayes' theorem (Bayes, 1763; Stigler, 1982) translates $P(X_{ni} = x|\underline{\theta}_n = \underline{\theta})$ the conditional probability of a particular response to a question given an examinee's proficiency, into $P(\underline{\theta}_n = \underline{\theta}|X_{ni} = x)$ the conditional probability of the examinee's proficiency given his or her question responses. Bayes' theorem expresses the information about an

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

examinee's proficiencies that is provided by question response $X_{ni} = x$ as follows:

$$P(\underline{\theta}_n = \underline{\theta} | X_{ni} = x) = \frac{P(X_{ni} = x | \underline{\theta}_n = \underline{\theta}) P(\underline{\theta}_n = \underline{\theta})}{P(X_{ni} = x)}.$$

For simplicity, in the equation above we have suppressed item parameter $\underline{\beta}_i$. The expression on the left $P(\underline{\theta}_n = \underline{\theta} | X_{ni} = x)$ is what we want to know, namely the probability that the examinee's proficiency value $\underline{\theta}_n = \underline{\theta}$ given his or her question response. In the numerator on the right, the first expression $P(X_{ni} = x | \underline{\theta}_n = \theta)$ is the item response model as we discuss in the section below on types of item response models. The second expression in the numerator on the right is called the prior probability of the proficiency variable $\underline{\theta}$, where "prior" means before taking into account the data $X_{ni} = x$. The denominator is a normalizing quantity that ensures that the expressions on the left add to 1.0 over all $\underline{\theta}$ (or integrate to 1.0 over the latent $\underline{\theta}$ space for continuous $\underline{\theta}$). That is required for the expression $P(\underline{\theta}_n = \underline{\theta} | X_{ni} = x)$ on the left to be a probability.

A similar expression can be written for $P(\underline{\theta}_n = \underline{\theta} | \underline{X}_n = \underline{x})$ where $\underline{X}_n = (X_1, \ldots, X_I)$ represents the vector of *all* scored item responses for a given student. In this more general case, the corresponding expression on the right hand side of Bayes' theorem can be expressed further in terms of the individual item response models $P(X_{ni} = x | \underline{\theta}_n = \underline{\theta})$ for each of the questions in the test. Thus Bayes' Theorem encapsulates the general evidence engine for an assessment and turns question performance $X_n = (x_1, \ldots, x_I)$ into inference about proficiencies $P(\underline{\theta}_n = \underline{\theta} | \underline{X}_n = \underline{x})$.

The Machinery of Measurement: Item Response Theory

Item response theory (IRT), in its broadest sense, serves as a general set of methods and approaches for quantitative modeling of assessment systems and analysis of assessment data (a number of classical references are available, including Baker, 2001; Crocker & Algina, 2006; Hambleton, Swaminathan, & Rogers, 1991). The general item response function is formulated as a probability model that expresses the probability of an examinee's response to a test question as a function of characteristics of the examinee and properties of the question:

$$P(X_{ni} = x | \underline{\theta}_n, \underline{\beta}_i).$$

Here $X_{ni} = x$ represents a particular scored response x to question i by examinee n; $\underline{\theta}_n$ represents one or more latent proficiencies of examinee n; and $\underline{\beta}_i$ consists of one or more parameters that represent properties of question i. For example, question i may be scored as $X_{ni} = 1$ for correct, and $X_{ni} = 0$ for incorrect. Underscores, as in $\underline{\theta}_n$ and $\underline{\beta}_i$, indicate that in general these variables can be vectors. Examinee proficiencies $\underline{\theta}_n$ can be continuous or discrete. The most common latent proficiency variable θ_n is a single real number, in which case we call the model a unidimensional, continuous model.

In expressing the probability of a question response as a function of examinee proficiency θ_n and question properties β_i, such a model postulates that examinee knowledge states and question parameters can be considered independently of any particular set of questions or fixed sample of students. This sample- and test-independence moves IRT beyond the more primitive ideas of classical test theory in which the basic unit of analysis is an administration of an intact test that consists of a fixed set of item responses from a particular sample of examinees. Classical test theory provides no theoretical base for inferring from results of a particular test to properties of a given question or subset of questions. If a subset of questions is taken from one test and added to a second test, a classical test theory analysis must be performed again from scratch. By contrast, IRT allows the model parameters of a given set of items to be estimated through a model calibration procedure from a representative set of data. Those items and their estimated model parameter values then can be used

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

individually and in groups without the need for recalibrating the question parameters. A dual property holds for examinees. Once an examinee's proficiency variables have been estimated through one set of questions, IRT allows us to predict that examinee's performance on other calibrated items from the same domain (see Crocker & Algina, 2006).

Types of IRT Measurement Models

To illustrate the measurement process we provide examples of two types of item response models, one continuous and the other a discrete model suitable for diagnostic assessment. A unidimensional, continuous latent proficiency variable θ is appropriate when the knowledge being measured can reasonably be thought of as a "level" along a continuum. This can be justified for relatively broad measurement domains, as in the SAT Mathematics test or the Graduate Record Exam subject matter tests in fields such as physics, mathematics, and computer science. Modeling on a continuous scale may be appropriate for end-of-year or midterm tests in a science or engineering courses. In these cases the test outcome is a "scale score" that represents a level of achievement along a continuum.

The first example is called the *2-parameter logistic (**2PL**) model:*

$$P(X_i = 1|\theta, a_i, b_i) = \frac{1}{1 + \exp(-Da_i(\theta - b_i))}.$$

Here θ is a continuous real number that represents *examinee ability level*, and there are two item parameters $\underline{\beta}_i = (a_i, b_i)$ for item i, where b_i is a real number called the *item difficulty parameter*, and a_i is a positive real number called the *item discrimination parameter*. The parameter D = 1.7 is a scaling factor used for technical reasons and is held fixed over all items. Both θ and b_i can be any real numbers, positive, zero, or negative, and a_i can be any positive real number (see Baker, 2001; Crocker & Algina, 2006; Hambleton et al., 1991).

An important special case of the 2PL model is the *Rasch model* or *1-parameter logistic (**1PL**) model*. This is defined by the same function as the 2PL model with the restriction that the discrimination values a_i must be the same for all items on a given test, for example, restricting all $a_i = 1$. A typical structural assumption on the examinee proficiency space is that the set of all population θ values is normally distributed with mean 0.0 and variance 1.0.

For any fixed item parameters a_i and b_i we can consider the 2PL item response model as a function of the proficiency variable θ :

$$f(\theta) = f_{a_i,b_i}(\theta) = P(X_i = 1|\theta, a_i, b_i).$$

We can see from the graphs of 2PL models shown in Figure 29.3 that the probability of a correct response increases as θ increases from a probability of nearly 0.0 on the left to nearly 1.0 on the right. Thus the model shows an increased probability of answering the question correctly as ability θ increases. The two item response functions with $a_i = 1$ are "parallel" across the entire latent space – that is, the curves are exactly the same shape and they do not cross. The curve with $a_i = 2$ increases more steeply at $\theta = 0.0$, and that curve crosses both of the other curves.

The 2PL item difficulty parameter b_i identifies the point $\theta = b_i$ on the θ scale at which the probability of a correct answer is 0.5. That is $f(b_i) = P(X_i = 1|\theta = b_i, a_i, b_i) = 0.5$. We can see from Figure 29.3 that as b_i increases, the curve moves to the right and the item becomes more difficult for any given θ level. For a fixed item difficulty value b_i, as the discrimination parameter a_i increases, the slope of the curve at that center point $\theta = b_i$ increases, and the item becomes easier for ability levels $\theta < b_i$ and more difficult for θ levels $\theta > b_i$. Under reasonable assumptions it can be shown that a_i is a monotonic function of the biserial correlation between item response and total score, and so is a measure of how well an item discriminates between high- and low-scoring examinees.

EXAMPLES OF DIAGNOSTIC MODELS

Diagnostic classification models (DiBello & Stout, 2007) are a type of item response

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

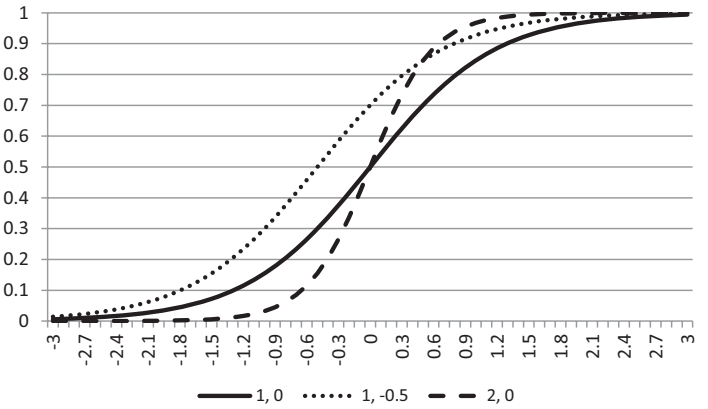


Figure 29.3. Graphs of three logistic item response functions. The legend gives the (a, b) values. The first two curves (solid and dotted lines) have parameters $(a, b) = (1, 0)$ and $(a, b) = (1, -0.5)$. The third curve (dashes) has $(a, b) = (2, 0)$. Since the first two curves have equal discrimination parameters (each has $a = 1$) they can be considered Rasch models with $b = 0$ and $b = -0.5$ respectively.

model in which the proficiency variable is a discrete profile of mastery levels for a set of aspects of knowledge and understanding:

$$P(X_i = x|\underline{\alpha}, \underline{\beta}_i).$$

Here the proficiency variable is a discrete vector $\underline{\alpha} = (\alpha_1, \ldots, \alpha_K)$ where each α_k represents an examinee's level of mastery of a particular facet k of knowledge or understanding. For example we could interpret $\alpha_k = 0, 1, 2$, as low, middle, or high proficiency in facet k knowledge; or, more commonly, α_k can be dichotomous, where $\alpha_k = 0$ or 1 represents nonmastery or mastery, respectively, of facet k knowledge. As part of its definition a diagnostic classification model includes an $I \times K$ Q-matrix of 0/1 elements, where I = number of items, K = number of identified facets of knowledge and understanding, and element $q_{ik} = 1$ or 0 indicates that item i either requires or does not require knowledge or understanding k.

An example of a diagnostic classification model is the fusion model (DiBello, Roussos, & Stout, 2007; DiBello & Stout, 2003, 2007) which is defined for item i as:

$$P(X_i = 1|\underline{\alpha}, \pi_i^*, r_{ik}^*) = \pi_i^* \prod_{k=1}^{K} (r_{ik}^*)^{q_{ik}(1-\alpha_k)}.$$

The latent space for the fusion model is the set of all alpha vectors $\underline{\alpha} = (\alpha_1, \ldots, \alpha_K)$ where each $\alpha_K = 0$ (nonmastery) or 1 (mastery), or other ordered levels. The fusion model expression is easy to parse. A student who is a master $\alpha_k = 1$ of each facet of knowledge or understanding that is required for a given item will have every exponent $q_{ik}(1 - \alpha_k) = 0$, and the fusion model expression reduces to π_i^*. That is the maximum possible value for the model probability. By contrast, if the student is a nonmaster for some k required by the item, then the exponent $q_{ik}(1 - \alpha_k) = 1$, and the probability of a correct answer includes multiplication by r_{ik}^*. Each r_{ik}^* is a number less than 1.0 and greater than or equal to 0.0, and acts as a multiplicative penalty that diminishes the probability of a correct response. The more nonmastered skills k there are and the smaller the values of r_{ik}^* the harsher is the penalty and the smaller the probability of a correct answer.

Diagnostic classification models are designed for cases when items require one or more than one skill. The fusion model makes a *conjunctive* assumption that high probability of answering an item correctly requires that the examinee be a master of *all* skills required for that item. Other diagnostic models are available with alternative

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

assumptions. The latent space of all possible $\underline{\alpha}$ vectors is finite. When each α_k is dichotomous, there are 2^K possible latent proficiency profiles $\underline{\alpha}$. The probability structure on the latent proficiency space can be any discrete probability distribution $p(\underline{\alpha})$ on the space of all 2^K $\underline{\alpha}$ vectors such that each $p(\underline{\alpha})$ is a nonnegative real number and the sum of $p(\underline{\alpha})$ over all $\underline{\alpha}$ is 1.0.

Reliability

TEST SCORE RELIABILITY

An important property of an assessment is the reliability of its score, traditionally discussed in the context of the observed total score T_{observed}. The reliability of measurement can be thought about from first principles in two different ways. First we can ask how closely correlated our observed total score T_{observed} is to the unobservable true score T_{true}. For technical reasons we define the squared value of this correlation as a measure of score reliability. A different way of thinking about reliability is to imagine that a given examinee takes a test once, his brain is wiped clean, and then he takes the same test again. If his observed total score is $T^{(1)}_{\text{observed}}$ from the first performance and $T^{(2)}_{\text{observed}}$ from the second, the second idea of reliability is how closely $T^{(1)}_{\text{observed}}$ and $T^{(2)}_{\text{observed}}$ are correlated. Under the assumptions of classical test theory it can be shown that

$$\left(\rho_{T_{\text{observed}},T_{\text{true}}}\right)^2 = \rho_{T^{(1)}_{\text{observed}},T^{(2)}_{\text{observed}}},$$

using $\rho_{X,Y}$ for the correlation between random variables X and Y. Thus the two notions are equivalent (Lord & Novick, 1968). Of course neither correlation can actually be computed. The true score T_{true} is not known and not observable, and a person's brain cannot be wiped clean. A variety of reliability indices are available that approximate reliability for a given test score, including Kuder–Richardson 20 (KR20) and Cronbach's alpha (see Baker, 2001; Crocker & Algina, 2006; Hambleton et al., 1991; van der Linden & Hambleton, 1997). In the framework of IRT the measurement outcome for a given examinee is his estimated proficiency level $\hat{\theta}$, and the analogous question is how reliable is the estimated proficiency level $\hat{\theta}$? We emphasize that the KR20 and Cronbach alpha indices for score reliability are appropriate in the context of IRT for the case when the proficiency variable is a total score or a unidimensional continuous latent ability $\hat{\theta}$.

RELIABILITY IN IRT MEASUREMENT MODELS

As discussed earlier, measurement plays a critical role in conceptualizing and formalizing aspects of inferential validity for an assessment. In particular, all model parameter estimates and proficiency level inferences produced by IRT are qualified by numerical measures of error or uncertainty. For example, for the unidimensional models 1PL or 2PL we are able to compute a conditional standard error of measurement for estimated proficiency $\hat{\theta}$ that effectively provides a 95% confidence interval around the estimate $\hat{\theta}$ of student latent ability. In the case of diagnostic classification models, discussed earlier, the proficiency variables are discrete and the reliability of diagnostic classifications is defined as an index of classification consistency. For example, in the case of a diagnostic assessment the mastery classification inference is about whether an examinee is a master or nonmaster of a particular skill, and we can ask the same two reliability questions from first principles as follows: What is the degree of consistency (1) between the true and inferred classifications, or (2) between the classifications inferred from administering the same test twice, wiping the brain between the two? The fundamental notions are the same for *any* IRT model, though the actual indices computed will differ from those used in unidimensional cases.

RELIABILITY CRITERIA

Educational measurement is imprecise to the extent that it yields best-possible inferences of unobservable latent proficiencies from observed student performance information. It is common for large-scale

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

standardized assessments to set high minimum acceptable reliability criteria, such as, for example, Cronbach's alpha should be greater than 0.90. In fact the needed level of reliability for a given assessment depends on the purpose and context of use of the assessment. In particular the requirement of very high reliability is largely moot for a classroom test whose use is primarily formative in providing learners and instructors with information about levels of knowledge and understanding of individual students or groups of students for purposes of instructional decision making and actions. For formative classroom assessment use it is important to determine the value of information provided by an assessment for purposes of deciding what kind of activity or classroom discussion to have next.

Putting the Pieces Together: Model Selection and Finding the Sweet Spot

The item response model most appropriate for a given assessment must be selected based on substantive domain assumptions. In the framework of the assessment triangle and evidence-centered design, the claims about students must be translated into concrete latent variables, either continuous or discrete, with one or more dimensions. Definitions of the student variables and their levels of proficiency must be chosen in such a way as to be able to translate freely between proficiency levels and claims about what students know and can do.

To apply the tools and methods of educational measurement to assessment means making the conceptual assessment framework concrete by (1) formally selecting and identifying the latent proficiency space as a probability space (the student model); (2) providing specific rules for turning observable evidence obtained from examinee performance on the selected assessment tasks into evidence that is usable by the item response model (which we will call the local evidence model); and (3) explicitly connecting the evidence thus obtained to updating the student model proficiency variables (called the global evidence model or the assessment scoring model). In complex cases such as those found for assessments intended for classroom use, the selection of an appropriate model can present a significant challenge. A suitable set of student model variables together with an appropriate item response model must satisfy the following three properties simultaneously:

1. The proficiency variables and the model must reflect the purpose and intent of the assessment design;
2. The proficiency variables must be aligned with both curricular materials and instructional practice; and
3. The skills and model must be statistically supportable.

These three properties are in tension with one another. For example, satisfaction of properties 1 and 2 calls for many relatively small-grained variables. Property 3 requires fewer variables for a given test length. Consequently the grain size will be coarser for at least some of these fewer variables. Searching for a set of variables and item response model that satisfy all three properties simultaneously can be thought of as chasing after an elusive "sweet spot." The latent variables and the item response model must satisfy all three properties to be able to achieve high statistical quality and high assessment validity.

Summary of Measurement Applied to Assessments in Engineering Education

The general class of models that fall under the category of item response models provides a broad set of modeling capabilities that can be applied to a wide range of engineering education settings. Selecting model type and details of the modeling is a necessary creative act that must occur before satisfactory analysis of assessment data can be accomplished. It can be considered as the mathematical and statistical formalization of the assessment triangle and evidence-centered design foundations as discussed earlier. In general, the item response modeling approach provides a broad set of

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

alternative models that can be fit to a wide range of assessment designs. Further, IRT comes equipped with a set of tools and methods for evaluating the performance of the model relative to the testing purpose by measuring errors of estimate and degree of data–model fit. Item response models help organize and interpret data, perform inferences about student proficiency relative to a particular student model and set of proficiency variables, and estimate the values of model parameters so that they can be used and interpreted.

Examples of Assessment and Measurement Applications in Engineering Education

There are many different types of assessment one might wish to develop and deploy for purposes of engineering education research. They can vary with respect to the constructs of interest (e.g., conceptual knowledge, problem solving, engineering design, interest, motivation) as well as the intended use (formative, summative, or program evaluation). It is beyond the scope of this chapter to consider each of these in depth. Rather, we illustrate how the ideas discussed in the first three main sections can be applied to an understanding and analysis of some particular examples drawn from research in engineering education.

To start our discussion of examples, it is useful to think about engineering knowledge, skills, and attributes in terms of two major claim spaces: (1) engineering knowledge and problem solving and (2) engineering professional skills. This distinction follows from a review of the engineering accreditation board's major criteria for evaluating engineering education programs. Five of the major ABET a–k criteria define knowledge and problem solving using descriptive action verbs fundamental to defining competency for the engineering tasks of design, troubleshooting and analysis (Brophy, Klein, Portsmore, & Rogers, 2008). ABET recommends engineering students be able to (1) design systems to meet needs (criterion c), which requires abilities to (2) identify, formulate, and solve engineering problems (criterion e); (3) apply mathematics, science, and engineering principles (criterion a); (4) use the techniques, skills, and tools of modern engineering (criterion k); and (5) design and conduct experiments and analyze and interpret data (criterion b). This ordering of criteria provides a sequence of knowledge and skills consistent with various models of problem solving in engineering (Woods, 1997, mathematics (Polya, 1945), and creative problem solving (Bransford & Stein, 1993. The specificity of the language associated with these criteria is an important step in the direction of clarifying the claims one would want to make about student competence and it makes the definition of observable and measurable performances easier to specify in terms of specific actions and products expected of learners.

The other six ABET criteria (d, f, g, h, i, j) identify important knowledge, skills, and attributes that students need to develop to be productive participants in their engineering professional careers. These criteria highlight the need to *function* on multidisciplinary teams, *communicate* effectively, *understand* the impact of engineering on society, *understand* professional and ethical responsibility, demonstrate *knowledge* of contemporary issues, and *recognize* the need for life-long learning. These criteria are admittedly broad and require refinement to articulate better the nature of the competencies desired as well as what one would need to observe in students' behavior to provide evidence of proficiency. Although discussion of measures of professional skills is outside the scope of this chapter, the design, development, and validation of assessments of such competencies would necessarily follow the logic presented earlier in the first three main sections.

In the remainder of this section we consider assessments that have proven to be of significant interest in STEM education relative to the general category of knowledge and problem solving. We start with the

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

case of concept inventories, then consider related examples of technology-based classroom formative assessment and end with a short discussion of problem-solving innovation. The theme throughout this section is how the assessment foundations provided by the assessment triangle and the evidence-centered design approach, combined with psychometric modeling and analysis, offer an interpretive framework for understanding the design and validity of an assessment relative to its intended purpose.

Concept Inventories in Engineering Education

A fundamental characteristic of expertise is the ability to reason with and apply disciplinary concepts to solve problems (e.g., Chi, Glaser, & Rees, 1982). Therefore it is not surprising that ABET lists this criterion first and engineering instructors make it a primary learning outcome for their courses. Engineers use their conceptual understanding of the sciences and the tools of engineering to comprehend and explain how systems work. These explanations are a critical step in troubleshooting existing systems, designing new systems, and constructing quantitative models to analyze these systems. Unfortunately, instructors in engineering programs regularly raise concerns about students' difficulties in developing deep conceptual understanding within their particular engineering disciplines and cite the challenges they face as instructors in assessing such understanding. Broad coverage of established sets of topics as commonly taught in engineering courses and a focus on developing practical problem-solving skill too often result in students able to pass course exams without achieving deep disciplinary understanding. Students are often unable to generalize their knowledge beyond the specific context of problems with which they are familiar. Concept inventories have been touted as one solution to this instructional and assessment dilemma and they are increasingly deployed in STEM courses as tools for both student and instructional evaluation.

Concept inventories provide good examples of the use of conceptual models of understanding in instructional domains to generate sets of test questions systematically (the cognition and observation vertices of the assessment triangle). In general, concept inventories are intended to measure students' deep conceptual understandings within a relatively narrow domain and they share a number of common features. Typically, they consist of multiple-choice questions whose distractor options were developed by first asking students open-ended questions and analyzing the student responses obtained. The questions are frequently based on science and engineering education research, including research on misconceptions and common student errors. Many inventories have developed explicit links between incorrect multiple-choice distractors and specific misconceptions. A wide array of inventories is now available in STEM education (http://ciHUB.org). For purposes of discussion, we use a particular example of a concept inventory to illustrate how issues of assessment design connect with issues of measurement, inference, and interpretation.

CONCEPT ASSESSMENT TOOL FOR STATICS

Concept Assessment Tool for Statics (CATS) is a concept inventory whose purpose is to diagnose statics concepts that engineering students typically have difficulty understanding and to identify persistent types of student errors (Steif & Dantzler, 2004; Steif & Hansen, 2006). The test consists of twenty-seven multiple-choice questions with three questions representing each of nine concepts: *Drawing Forces on Separated Bodies (Free Body Diagrams)*, *Newton's Third Law*, *Static Equivalence*, *Roller Joint*, *Pin-in-Slot Joint*, *Loads at Surfaces with Negligible Friction*, *Representing Loads at Connections*, *Limits on Friction Force*, and *Equilibrium* (Steif & Hansen, 2007). Each question was designed to test one of the nine concepts in isolation and to require little or no calculation. Wrong answers were designed to reflect known student errors.

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

Table 29.1. Cronbach's α for CATS Subscales

Concept	α	N
Free body diagram	.72	3
Newton's third law	.71	3
Static equivalence	.61	3
Roller	.68	3
Slot	.68	3
Frictionless point contact	.47	3
Representation	.33	3
Friction	.64	3
Equilibrium	.37	3

CATS is available online through the website http://ciHUB.org for access by instructors and for administration to students and is used regularly by numerous instructors. Instructor reports give item right and wrong, total score, and student concept scores (0, 1, 2, or 3 correct of the three questions for that concept). How answers connect to the specific errors and misconceptions linked to wrong answer choices are not part of the typical reporting process. Paul Steif, the developer of CATS, was kind enough to provide us with 1,372 cases of CATS student response data that had been collected online during a one-year period. We were able to analyze those data to generate evidence about CATS given its purpose, design, and scores. We frame our analysis in terms of a set of questions one would want to ask about the reliability and validity of any given concept inventory or similar type of assessment.

Some Basic Item and Test Statistics. One question that is important to justify use of the CATS overall total score is its reliability (see definition of reliability in the section "Issues of Measurement: Interpretation and Statistical Inference"). Our analyses reveal that the CATS total score is highly reliable (twenty-seven items; $\alpha = 0.84$). Given that scores also are reported for the nine separate concept groups we also calculated reliabilities for the subscales. All but two demonstrated reasonable reliabilities for a three-item scale as shown in Table 29.1.

A second set of issues important for interpretation of score outcomes relates to how well the individual items function as probes of student proficiency in this domain of thinking. Individual item difficulties can be defined as the proportion of students answering a given item correctly. For CATS these ranged from 0.16 to 0.78, and, with the exception of question 26 at 0.16, most would consider a range of item difficulties from about 0.20 to 0.80 to be reasonable. A spread of item difficulties supports the process of discriminating among examinees who vary in their proficiencies with respect to this domain of knowledge and skill. A second characteristic of item functioning is item discrimination, which is the biserial correlation between each person's score on a given item and his or her total score on the rest of the instrument. For CATS the values ranged from 0.18 to 0.49. In essence, an item's discrimination value is a measure of its sensitivity to the student's overall ability, so that those who do well on the overall assessment also tend to do well on the individual item. Again, except for Q26 with discrimination 0.18, the values obtained for the CATS items constitute reasonable evidence that each item is generally positively related to the overall proficiency represented by the aggregate of performance on the rest of instrument.

Application of Unidimensional IRT to CATS. As noted earlier, IRT provides a powerful method for evaluating the functioning of an assessment. For CATS, the model-data fit indices of both 1PL and 2PL models were good. One interpretive advantage of an IRT analysis is that it allows us to place the items and examinees in the same space. An example is the "Wright map" shown in Figure 29.4. The vertical axis represents a scale for the "latent CATS ability" θ, which runs in the figure from -3.0 to $+3.0$. Indicated to the left of the axis are the locations of each student's θ and the distribution of those scores across the total examinee sample. The abilities are scaled to have a mean of 0.0 and a standard deviation of 1.0.

The twenty-seven questions numbered Q1 through Q27 are shown to the right of the central axis. Each question is located at the scale value of its b_i 1PL item difficulty

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

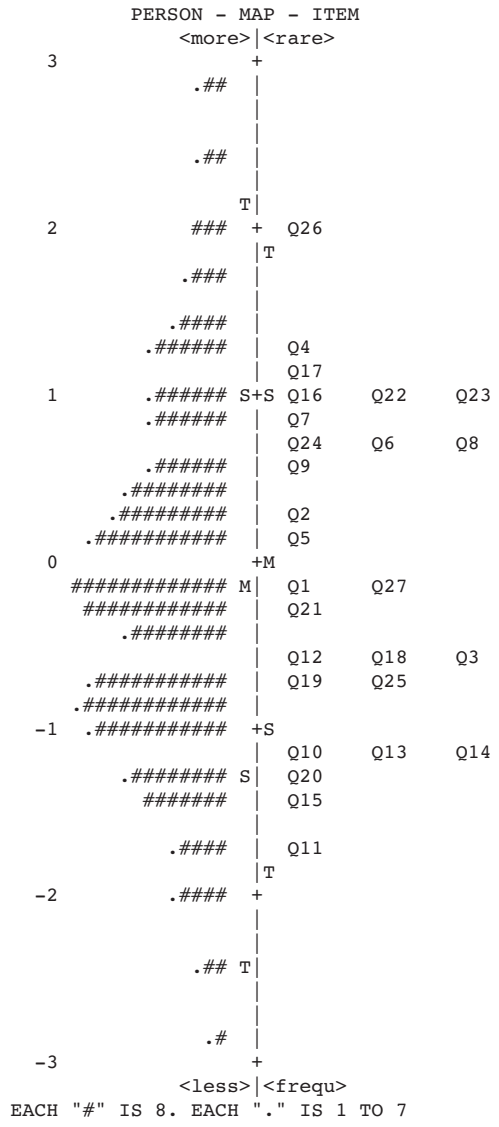


Figure 29.4. Wright map of CATS data. The scale on the central axis goes from −3.0 at the bottom to +3.0 at the top. The symbols on the left stand for location of examinee abilities θ, with "#" representing eight examinees and "*" representing one to seven examinees. Mean of examinee θ is 0.0. The right side of the axis shows item numbers Q1 through Q27. Each item number is located at the 1PL difficulty parameter b_i.

parameter (see earlier discussion in the section "Issues of Measurement: Interpretation and Statistical Inference"). Recall that the b_i item difficulty parameters are monotonically related to the proportions of examinees answering an item correctly. Thus items located at low points on the scale (such as Q11, Q16, and Q20, for example) are relatively easy, and items located at higher levels of the scale (like Q26 [the most difficult item], Q4, and Q17) are more difficult.

Placing the questions and examinees on the same scale allows us to see whether the question difficulties and examinee abilities are well matched. Notice that there is a set of about 5% of the students with estimated abilities −2.0 and below, and there are no comparable questions with difficulty parameters in such a low range. After the most difficult and problematic question (Q26), the next most difficult question was question Q4 with difficulty about 1.25, and a comparable set of about 10% of students have ability levels above 1.3. In general, the set of questions seems reasonably well matched to this range of student abilities.

Structural Analysis of the Data. An important empirical test of the developers' claims about questions clustered by concept is the structural consistency between the data and the developers' concepts. We performed a series of exploratory factor analyses (EFAs) to determine the best number of factors supported by the data, and the degree of match or mismatch between the factor loadings of questions and the concepts to which the question are assigned. An EFA with varimax rotation identified nine factors that collectively explained 57.6% of the variance. These results generally supported the developer's a priori nine clusters and indicated that the assessment is tapping important and differentiable aspects of student knowledge and understanding in the domain of statics.

We also performed confirmatory factor analyses (CFAs) to examine causal effects of the developers' concepts or combinations of concepts on question performance and associations among concepts. The best fitting model (see Figure 29.5 for a representation of the estimated model) was a seven-factor model that omitted problematic question 26 and combined two pairs of the hypothesized nine categories: Equivalence and Equilibrium and Frictionless and

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

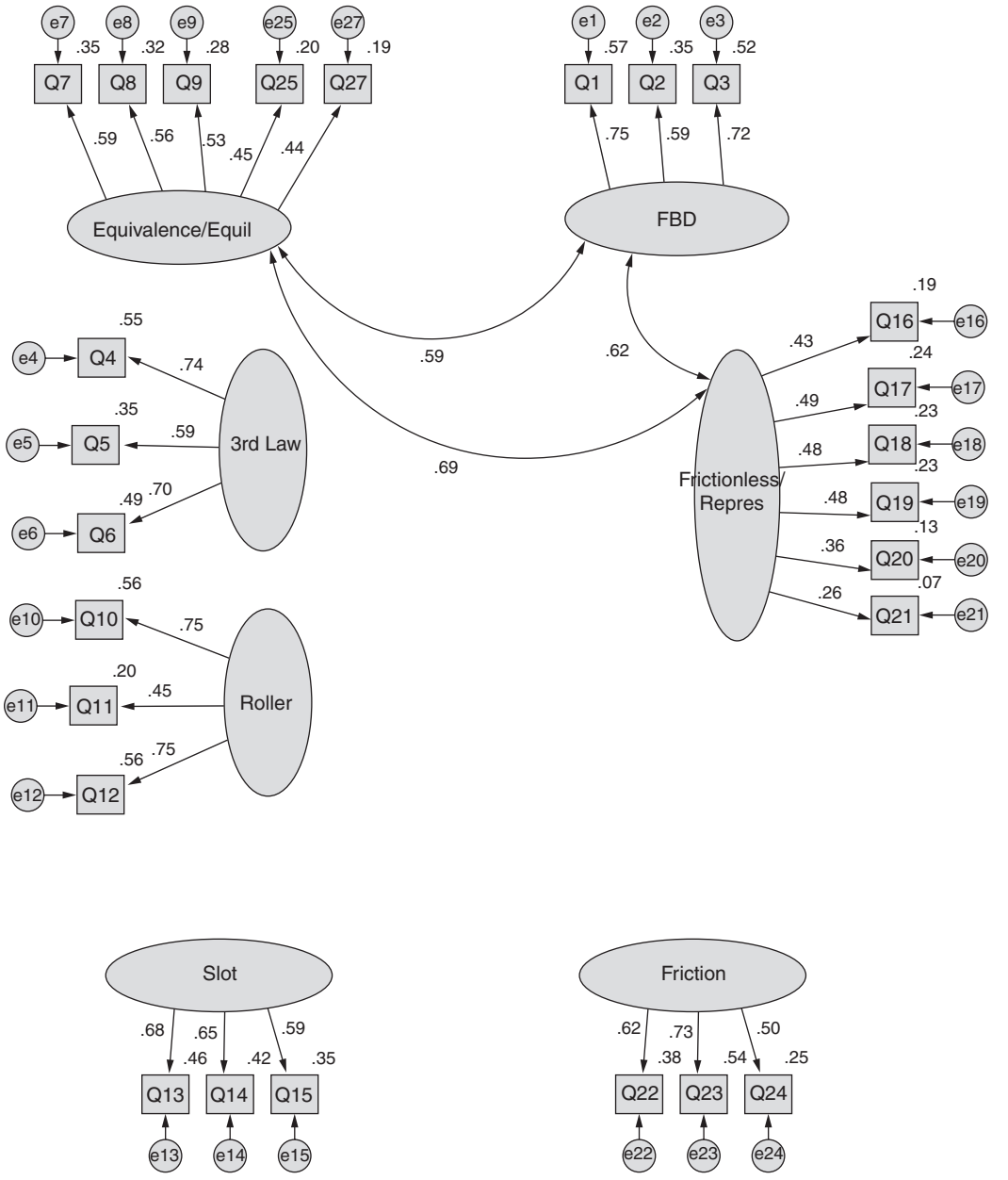


Figure 29.5. Final seven-factor CFA model for CATS.

Representation. This model included correlations among three latent constructs: Free Body Diagrams, Equivalence/Equilibrium, and Frictionless/Representation.

We also examined the diagnostic possibilities of the CATS through the complementary lens of a fusion model analysis (Santiago-Román, 2009). The CATS instrument showed strong diagnostic strength relative to identified facets of knowledge and understanding. A set of ten skills was developed that consisted of some original concepts unchanged and some new skills made up of combinations of questions from several of the original concepts. Items were freely coded as requiring one or more than one of the defined skills. The ten skills were identified through an iterative sequence of

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

substantive modifications to the current skills and Q matrix followed by diagnostic model calibration runs. Model–data fit measures strongly supported the set of ten skills. Details can be found in Santiago-Román (2009), Santiago-Román, Streveler, and DiBello (2010) and Santiago Román, Streveler, Steif, & DiBello (2010). The diagnostic analysis showed a striking concordance with the seven-factor best fitting confirmatory factor analysis model shown in Figure 29.5. All of the diagnostic skill definitions that were different from the original nine concepts were confined within the correlated portion of the model shown in Figure 29.5, namely the three latent variables identified as Equivalence/Equilibrium, Free Body Diagrams, and Frictionless/Representation.

The structural analyses taken together – the EFA, CFA, and diagnostic analyses – shed light on the functioning of CATS relative to its design and suggest possibilities for formative classroom use. The analyses offer an interpretive argument that links the CATS substantive foundations with the correlational structure found in the data as represented by Figure 29.5. Just as important they provide a reasonably strong warrant for the formative classroom use of diagnostic outcomes from CATS in addition to the existing and more typical use of summative scores.

APPLICABILITY OF THE APPROACH TO ALL CONCEPT INVENTORIES

In light of the rich conceptual and cognitive models guiding concept inventory (CI) item development, the analytic framework for assessments that is made up of the assessment triangle and the evidence-centered design approach, along with the application of a reasoned and principled combination of more powerful measurement models, makes possible a more extended understanding and appreciation of the inferential linkage from observation to interpretation and from interpretation back to cognition. The methods exemplified here constitute a good start at confirming the strong interpretive frameworks through more sophisticated psychometric techniques. They can help capture the possible diagnostic power of instruments such as CIs, and enhance their usefulness as both summative and formative assessment tools. Hopefully, this conversation about methods and approaches encourages engineering educators to raise the bar for the validity and quality of assessments for use in STEM classrooms and heightens an appreciation for what constitutes sufficient conceptual and empirical support for assessment claims.

Technology and Classroom Formative Assessment Processes

The issues highlighted in the preceding discussion regarding student difficulties with conceptual knowledge and understanding have resulted in many examples of attempts to design active learning environments in engineering education. The goal is to provide effective methods to engage learners in the classroom and overcome some of the conceptual challenges that students otherwise face (e.g., Prince, 2004; Roselli & Brophy, 2006; Smith, Johnson, Johnson, & Sheppard, 2005). Many of these designs rely on the use of classroom based formative assessment processes that begin by posing concept-based questions to students (e.g., Crouch & Mazur, 2001). Similar to questions found on concept inventories, these probes of student thinking are designed to require qualitative reasoning with key concepts to either explain or predict outcomes. Students are asked to generate a response (e.g., select a multiple-choice option; generate code, a diagram, written explanation, or a graph) and then share it with the instructor using various technologies. A very popular method uses Classroom Response System (CRS) technology to poll students' responses (e.g., Beatty & Gerace, 2009). Student responses are sent with wireless devices and aggregated into a summary chart at the instructor's computer. An instructor can use the results to determine how well students have comprehended key concepts. An instructional benefit occurs when a substantial portion of the students select

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036



Figure 29.6. Athlete in an iron cross position.

responses that represent one or more common misunderstandings. The instructor then can use a number of pedagogical approaches to provide feedback to students. It must be noted, however, that engaging in these activities requires very thoughtful item design to ensure that the desired range of performance outcomes is likely to occur.

For example, both the Force Concept Inventory (Hestenes, Wells, & Swackhamer, 1992) and CATS (Steif & Hansen, 2007) target multiple concepts associated with analyzing a system in static equilibrium (i.e., all forces balance, resulting in a motionless system). A critical step in this analysis is representing the forces acting on the system using a free body diagram (FBD) and generating Newton's laws that define the behavior of these forces. A course in biomechanics included this kind of analysis as part of its learning outcomes (Roselli & Brophy, 2003, 2006). A CRS system was used in the course to support development of students' ability to generate appropriate FBD representations (Barr, Cone, Roselli, & Brophy, 2003; Roselli & Brophy, 2003, 2006). The course used challenge-based instruction to engage students in a design task to develop practice equipment for novice athletes to perform the iron cross position with rings shown in Figure 29.6. The instructor engaged learners in an inquiry process to investigate various sub-problems involved in this subtask. As part of a lecture session the instructor posed the subtask shown in Figure 29.7 that asked students to generate a free body diagram of the athlete's hand and ring assembly. After a few minutes the students were then shown possible solutions (Figure 29.8) and asked to pick the one that best matches their solution. Students used a small remote device to make their selection and a frequency plot of all the responses was displayed.

At this point an instructor has a range of pedagogical options for providing feedback. The first is simply to state the correct

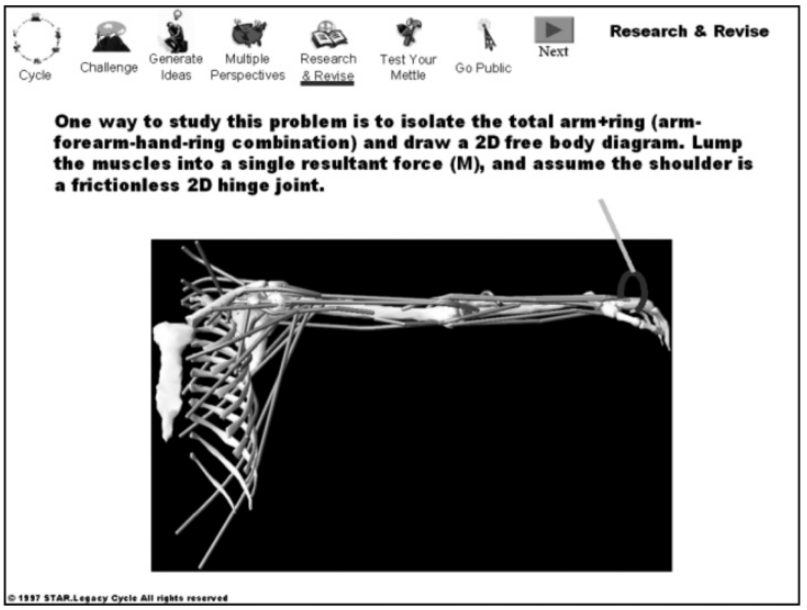


Figure 29.7. Pose problem and ask students to generate a solution.

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

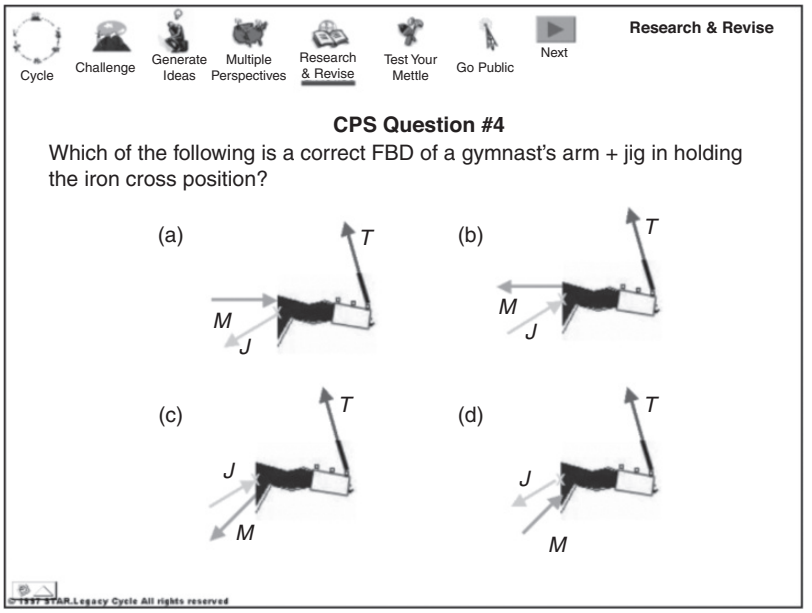


Figure 29.8. Students respond to a question based on what they constructed.

answer and move on. Students who answered correctly receive confirmation and will assume they reasoned correctly. Those who answered incorrectly know they need to refine their thinking, but have no additional information to help them do so. The instructor's second option is to work through the correct solution. This better assists students who chose the correct answer because it confirms the appropriateness of their reasoning about the problem. Students selecting one of the foils will need to recognize their errors on their own. Unfortunately, the worked out example may not address the learner's specific misconception or provide additional information to refine his or her thinking. The instructor's third choice would be to open a dialog with students to have them justify their responses. She could start with the most popular response first. If this response is not the correct answer, then the discussion can be quite productive because students will hear the different ways their peers conceive of the concepts underlying the problem. The instructor can repeat the process for each of the answer options as a way to review systematically all the ways in which members of the class may have been thinking about the problem situation. This approach provides a level of individual feedback that addresses the needs of all students, and research has illustrated its positive impact on overall student learning (e.g., Roselli & Brophy, 2006).

Although using a CRS is efficient, answering multiple-choice questions does not necessarily require students to reflect on their own work. The earlier example asked students to generate a diagram first and then evaluate four other solutions. Learners could choose to ignore what they generated and try to work backwards from the solutions provided. Therefore, an alternative to multiple choice would be an interactive system in which students could generate a FBD and then receive feedback on what they generated. Figure 29.9 illustrates an interactive software program that presents learners with a sequence of statics problems requiring analysis (Roselli, Howard, & Brophy, 2006). Students can insert vectors and moments to represent their analysis of forces on the system. Feedback is generated once they submit their answer. They can refine their answer and resubmit for additional feedback. The feedback is designed to provide increasing levels of specificity.

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

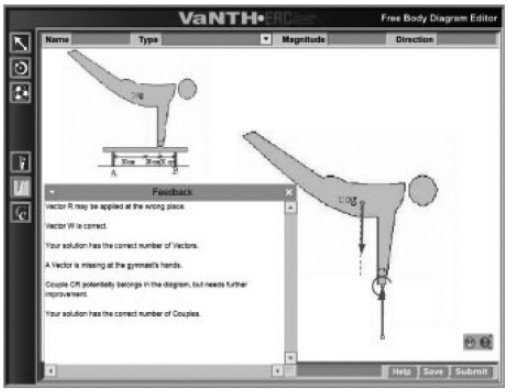


Figure 29.9. Free body diagram assistant.

The first round of feedback simply evaluates what is missing or potentially inappropriate. The second round of information is more specific about what is missing, such as the location or direction of a vector or moment. After three attempts students are provided the final solution for the diagram. Students using this system demonstrate an increase in their ability to generate diagrams for the specific joint types (e.g., contact, pin, or clamp). For example, with each new problem students required fewer submissions to achieve the correct solution. Systems like these have the potential of engaging students in problem-solving activities that replicate the professional experiences they need for engineering practice, and the systems provide mechanisms to help students develop the conceptual understanding they will need to be flexible and adaptive. Consistent with what we have argued in this chapter, designing such systems requires a very careful analysis of the claims one wishes to make about student competence and then designing ways to obtain relevant forms of evidence for either formative or summative assessment purposes.

Assessing Innovation and Adaptiveness in Problem Solving

Engineers continually generate new ideas designed to improve existing devices and systems (Dym et al., 2005). This kind of expertise involves skills associated with problem finding, identification, and formulation (Atman, Kilgore, & McKenna, 2008). As with other aspects of competence, assessment requires clarification of the knowledge, skills, and attributes associated with these various initial phases of problem solving. A related challenge is identifying the forms of evidence needed to support claims that students possess the relevant skills, as well as the tasks or environments that will elicit the relevant evidence. Given the importance of these competencies in engineering, it is not surprising that engineering education continues to expand its use of problem-based learning, which frequently relies on open-ended problems and projects that are presumed to tap the requisite knowledge and skills. Such problems may require weeks to solve and students often work in teams, making individual student evaluation a challenge.

Several approaches to measuring problem identification have focused on students' ability to evaluate novel problems using their domain knowledge and engineering tools to generate ideas, questions, and plans to solve a problem. This ability to adapt to novel problems illustrates how experts can transfer what they know into new contexts. This idea of adaptive expertise (Schwartz, Bransford, & Sears, 2005) highlights experts' ability both to solve familiar problems efficiently and systematically generate new knowledge to solve new problems. These two dimensions of expertise help instructors design differential learning objectives for their students. The first dimension is to rapidly and effortlessly solve common problems found in the target domain area. Experts' conceptual understanding of the domain can be used to solve routine problems with great speed and accuracy. The second dimension illustrates what occurs when an expert faces an unfamiliar problem. Though their speed and efficiency to approach the problem may be diminished, experts with the appropriate knowledge, skills, and attitudes engage in an inquiry process to generate the knowledge needed to identify possible solutions. This requires noticing what knowledge they don't know, generating questions they need to

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

answer, and formulating a plan and a process to find information and synthesize it into new knowledge. This requires looking beyond their normal perspective and managing uncertainty. Therefore, experts also display abilities to persevere and a willingness to deal with ambiguity. Note that the display of efficiency for applying conceptual knowledge is not always an indicator of adaptiveness (Hatono & Inagaki, 1986), or of the ability to think innovatively. The design of instruction and assessment requires a framework that better illustrates the relationships between efficiency and innovation and how the latter develops in learners (Schwartz et al., 2005). New measures are being explored to better assess engineering students' development of adaptiveness.

For example, a scenario-based problem method was used to assess student adaptiveness. Researchers were interested in measuring the development of efficient use of domain knowledge and the growth in adaptiveness (innovation) over time. Also, they wanted to explore the development of these characteristics relative to students' confidence. The method consisted of a pre–post assessment using a design (analysis) scenario. The assessment method presented students with a cardiologist's design for an implantable defibrillator that he postulated would redefine the industry's approach to the device. To measure efficiency ability, or their ability to generate an appropriate response, the students were asked "What do you need to do to test the doctor's hypothesis?" Second, to measure students' adaptiveness they were asked "What questions do you have for the doctor?" Finally students were asked "How confident are you in your response?" Each question for the doctor posed by the student was scored with a rubric developed with the assistance of domain experts in cardiology and the design process. The responses to the first question about testing the hypothesis were coded as frequency of strategies generated and the quality of the responses (e.g., 1, novice; 2, expert). The difference in the quality of questions posed by students was based on whether students' strategies were vague (e.g., gather data at sites) versus providing viable alternatives for testing the devices performance to prove feasibility (compare data at two sites to determine if both are necessary). The second question that asked students to generate questions for the doctor was coded by counting the number of questions asked and the quality of the questions (1, novice; 2, proficient; 3, expert). Novice questions focused more on factual information, like definitions. This included questions specific to the immediate problem (e.g., "Could I see tracing of the ECG?"). "Problem-specific knowledge" was coded as proficient. For example, expert questions focus more on understanding the rationale of the proposal and understanding of the range of alternatives already considered. These rubrics provided a quantitative score for the number and quality of students' responses.

This pre–post assessment was administered to both seniors in a capstone design class and first-year engineering students studying cardiac signal analysis in an introductory course. Seniors were better at defining an approach for testing the doctor's hypothesis and generated more questions to ask the doctor that were more like what an expert would ask. Seniors were also more confident than first-year students, but even first-year students showed an increase in these measures across their first year. This measure of adaptiveness illustrated that first-year students had a significant increase over the semester on both the efficiency and innovation scales. This kind of assessment could also be used to illustrate the growth of students' proficiency across the engineering curriculum (Walker, Cordray, King, & Brophy, 2006).

Assessing students' innovation skills (problem identification and formulation) can also be integrated into standard methods at the course level. Exams might be designed with various levels of questions. The first level of test items might focus on assessing students' efficiency defined in terms of basic taxonomic knowledge associated with facts and simple procedures. These items measure students' ability to apply fluently the

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

concepts and procedures they have practiced throughout the semester. Innovation questions are designed to evaluate students' ability to formulate a plan of action for a larger open-ended problem. The goal is not to generate and execute a final solution, but to formulate the problem appropriately by defining problems to be explored and questions to ask to acquire needed information. The third level of questions would focus on transfer and would consist of questions that are conceptually similar to problems in the course, but within a different context. These are adaptive expertise question items. The approach is similar to the scenario-based instrument described earlier, but the addition of specific targeted questions at differing levels can make for a richer assessment of what students know and can do. A version of this approach was explored in a third year biotransport course using a challenge-based instructional approach. The course had a strong focus on developing students' problem solving abilities while learning the constitutive properties (taxonomic knowledge) that govern the transport behavior of biological systems. The course was organized around three major types of biotransport: fluid, heat, and mass transfer. An exam was given after instruction via a series of challenge-based modules. Analysis of performance on three exams distributed across the semester illustrated that students increased their abilities for all three areas of competency. The efficiency and innovation skills developed quickly from exam 1 to 2, but adaptiveness was relatively flat over that same period. However, by the end of the semester all three dimensions had increased significantly (Martin, Petrosino, Rivale, & Diller, 2006).

As new instructional approaches make their way into the engineering education classroom, additional assessment methods will need to be designed and validated to evaluate student learning outcomes. Some efforts have been made to measure dimensions of engineering thinking and the generalizability of such approaches (Atman et al., 2008; Cordray, Harris, & Klein, 2009; Prince & Felder, 2006). Assessment methods must be sought that can demonstrate that students are developing both the problem-solving skills they need to solve routine problems and the executive skills needed for managing the process of investigating new problems. The previous examples illustrated text-based assessments of students' ability to articulate what they notice and what knowledge they need to generate. The next steps of measurement will involve students' ability to model systems to facilitate their comprehension and analysis of complex systems (Brophy & Li, 2011). A model-based reasoning approach (e.g., Nersessian, 2009) can involve qualitative reasoning with diagrams and graphs and analytic modeling to evaluate the behavior of systems quantitatively. These graphical images and how learners reason with them to generate new knowledge and insights makes engineering an interesting domain to rethink what and how we assess. Another dimension of engineering involves the decisions engineers make once they have evidence from their models and analysis. Therefore, other areas of investigation for assessment design can be in decision theory and choice (Schwartz & Arena, 2009). The challenge will be to create new measures that are valid, reliable, feasible, and useful to support students' learning and that can enhance the quality of an engineering education program.

Final Thoughts: The Role of Assessment in a Comprehensive R&D Program

High-quality evidence that permits instructors, researchers, and administrators to ask and answer critical questions about the outcomes of learning and instruction – what students know and are able to do – is critical to advancing an R&D agenda in engineering education. The first requirement for developing quality assessments is that the concepts and skills that signal progress toward mastery of a domain be understood and specified. These activities constitute fundamental components of a prospective R&D agenda as applied to multiple engineering

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

courses and content domains. Assessment of the overall outcomes of instruction is important to the R&D agenda because it allows for tests of program effectiveness. But it is more broadly important because the content of such assessments can drive instructional practice for better or for worse. The deployment of the force concept inventory in physics illustrates a case in which an assessment tool based on cognitive and instructional research has had a powerful, positive impact on the redesign of instruction, serving as an evaluation tool to determine the effectiveness of new instructional methods (e.g., Hake, 1998). Assessment of the impact of long-term programs of R&D is also important. University faculty, administrators, and alumni need ways to determine the return on investing in research in engineering education. Such a use of assessment for program evaluation purposes needs measures that are sensitive to learning and the impact of quality teaching. Researchers, educators, and administrators must therefore concern themselves with supporting research and development on effective and appropriate assessment procedures that can serve multiple purposes and functions as part of a coordinated system of assessments.

Although we have distinguished among three purposes of assessment, quality instruments for each depend on the same three components: (1) theories and data about content-based cognition that indicate the knowledge and skills that should be assessed; (2) tasks and observations that can provide information on whether students have mastered the knowledge and skills of interest; and (3) qualitative and quantitative techniques for scoring student performance that capture fairly the differences in knowledge and skill among students being assessed. Research is clearly needed in all three areas of work, as well as their integration.

Research also needs to explore (1) how new forms of assessment can be made accessible for instructors, and practical for use in classrooms; (2) how they can be made efficient for use in university teaching contexts; and (3) how various new forms of assessment affect student learning, instructor practice, and educational decision making. Research must explore ways that instructors can be assisted in integrating new forms of assessment into their instructional practices and how they can best make use of information from such assessments. Also to be studied are ways that structural features of instructional delivery in higher education (e.g., length of class time, class size and organization, and opportunities for students and/or instructors to work together) impact the feasibility of implementing new types of assessments and their effectiveness.

As a field, engineering education continues to make progress. Hopefully this chapter's discussion of ways to think about the design and uses of assessment provides a useful set of ideas and approaches that can further advance the field of engineering education research.

Acknowledgments

Preparation of this chapter was supported by the following grants from the U.S. National Science Foundation to James W. Pellegrino and Louis V. DiBello: CCLI Project No. 0920242, REESE Project No. 0918552, and REESE Project No. 0815065 and to Sean Brophy: DUE-0817486.

References

American Educational Research Association, American Psychological Association, & National Council on Measurement in Education (AERA/APA/NCME). (1999). *Standards for educational and psychological testing*. Washington, DC: Author.

Atman, C. J., Kilgore, D., & McKenna, A. (2008). Characterizing design learning: A mixed-methods study of engineering designers' use of language. *Journal of Engineering Education*, 97(3), 309–326.

Baker, F. (2001). *The basics of item response theory*. ERIC Clearinghouse on Assessment and Evaluation, University of Maryland, College Park, MD. Retrieved from http://echo.edres.org:8080/irt/baker/http://edres.org/irt/baker/

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

Barr, R. W., Cone, J., Roselli, R. J., & Brophy, S. P. (2003). Initial experiences using an interactive classroom participation system (CPS) for presenting the iron cross biomechanics module. In *Proceedings of the 2003 ASEE Gulf-Southwest Annual Conference*, the University of Texas at Arlington. Lubbock, TX.

Bayes, T. P. (1763). An essay towards solving a problem in the doctrine of chances. *Philosophical Transactions of the Royal Society of London*, *53*, 370–418.

Beatty, I. D., & Gerace, W. J. (2009). Technology-enhanced formative assessment: A research-based pedagogy for teaching science with classroom response technology. *Journal of Science Education and Technology*, *18*(2), 1146–1162.

Bransford, J. D., & Stein, B. S. (1993). *The IDEAL problem solver* (2nd ed.). New York, NY: W. H. Freeman.

Brophy, S., Klein, S., Portsmore, M., & Rogers, C. (2008). Advancing engineering in P-12 classrooms. *Journal of Engineering Education*, *97*(3), 369–387.

Brophy, S., & Li, S. (2011). Problem definition in design by first year engineering students. In *Proceedings of the American Society of Engineering Education Annual Conference*, Vancouver, BC, Canada.

Chi, M. T. H., Glaser, R., & Rees, E. (1982). Expertise in problem solving. In R. Sternberg (Ed.), *Advances in the psychology of human intelligence* (Vol. 1, pp. 7–76). Hillsdale, NJ: Lawrence Erlbaum.

Cordray, D. S., Harris, T. R., & Klein, S. (2009). A research synthesis of the effectiveness, replicability, and generality of the VaNTH challenge-based instructional models in bioengineering. *Journal of Engineering Education*, *98*, 335–348.

Crocker, L., & Algina, J. (2006). *Introduction to classical and modern test theory*. New York, NY: Wadsworth.

Crouch, C. H., & Mazur, E. (2001). Peer instruction: Ten years of experience and results. *American Journal of Physics*, *69*, 970–977.

DiBello, L. V., Roussos, L. A., & Stout, W. F. (2007). Review of cognitively diagnostic assessment and a summary of psychometric models. In C. R. Rao & S. Sinharay (Eds.), *Handbook of statistics*, Vol. 26: *Psychometrics* (pp. 979–1030). Amsterdam: Elsevier.

DiBello, L., & Stout, W. F. (2003). Student profile scoring methods for informative assessment. In H. Yanai, A. Okada, K. Shigemasu, & J. J. Meulmann (Eds.), *New developments in psychometrics* (pp. 81–92). Tokyo: Springer.

DiBello, L. V., & Stout, W. F. (Eds.). (2007) Special issue on IRT-based cognitive diagnostic models and related methods, *Journal of Educational Measurement*, *44*(4), 285–292.

Dym, C. L., Agogino, A. M., Eris, V., Frey, D., & Leifer, L. J. (2005). Engineering design, thinking, teaching and learning. *Journal of Engineering Education*, *94*(1), 103–120.

Haertel, E. H., & Lorie, W. A. (2004). Validating standards-based test score interpretations. *Measurement: Interdisciplinary Research and Perspectives*, *2*(2), 61–103.

Hake, R. R. (1998). Interactive-engagement versus traditional methods: A six-thousand-student survey of mechanics test data for introductory physics courses. *American Journal of Physics*, *66*(1), 64–74.

Hambleton, R. K., Swaminathan, H., & Rogers, H. J. (1991). *Fundamentals of item response theory*. Measurement Methods for the Social Science. Newbury Park, CA: SAGE.

Hatano, G., & Inagaki, K. (1986). Two courses of expertise. In H. Stevenson, H. Azuma, & K. Hakuta (Eds.), *Child development and education in Japan*. New York, NY: W. H. Freeman.

Hestenes, D., Wells, M., & Swackhamer, G. (1992, March). Force concept inventory. *The Physics Teacher*, *30*, 159–166.

Hickey, D., & Pellegrino, J. W. (2005). Theory, level, and function: Three dimensions for understanding transfer and student assessment. In J. P. Mestre (Ed.), *Transfer of learning from a modern multidisciplinary perspective* (pp. 251–293). Greenwich, CT: Information Age.

Kane, M. T. (2006). Validation. In R. L. Brennan (Ed.), *Educational measurement* (4th ed., pp. 17–64). Westport, CT: Praeger.

Lord, R. M., & Novick, M. R. (1968). *Statistical theories of mental test scores*. Reading, MA: Addison-Wesley.

Marion, S., & Pellegrino, J. W. (2006, Winter). A validity framework for evaluating the technical quality of alternate assessments. *Educational Measurement: Issues and Practice*, 47–57.

Martin, T., Petrosino, A. J., Rivale, S., & Diller, K. R. (2006). The development of adaptive expertise in biotransport. *New Directions for Teaching and Learning*, *108*, 35–47.

Messick, S. (1994). The interplay of evidence and consequences in the validation of performance

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

assessments. *Educational Researcher*, 23(2), 13–23.

Mislevy, R. J., & Haertel, G. (2006). Implications of evidence-centered design for educational assessment. *Educational Measurement: Issues and Practice*, 25, 6–20.

Mislevy, R. J., & Riconscente, M. M. (2006). Evidence-centered assessment design: Layers, concepts, and terminology. In S. Downing & T. Haladyna (Eds.), *Handbook of test development* (pp. 61–90). Mahwah, NJ: Lawrence Erlbaum.

Mislevy, R. J., Steinberg, L., & Almond, R. (2003). On the structure of educational assessments. *Measurement: Interdisciplinary Research and Perspectives*, 1, 3–67.

National Research Council. (2003). *Assessment in support of learning and instruction: Bridging the gap between large-scale and classroom assessment*. Washington, DC: The National Academies Press.

Nersessian, N. J. (2009). How do engineering scientists think? Model-based simulation in biomedical engineering laboratories. *Topics in Cognitive Science*, 1, 730–757.

Pellegrino, J. W., Chudowsky, N., & Glaser, R. (Eds.). (2001). *Knowing what students know: The science and design of educational assessment*. Washington, DC: The National Academies Press.

Pellegrino, J. W., & Hickey, D. (2006). Educational assessment: Towards better alignment between theory and practice. In L. Verschaffel, F. Dochy, M. Boekaerts, & S. Vosniadou (Eds.), *Instructional psychology: Past, present and future trends. Sixteen essays in honour of Erik De Corte* (pp. 169–189). Oxford: Elsevier.

Polya, G. (1945). *How to solve it*. Princeton, NJ: Princeton University Press.

Prince, M. (2004). Does active learning work? A review of the research. *Journal of Engineering Education*, 93(3), 223–231.

Prince, M. J., & Felder, R. M. (2006). Inductive teaching and learning methods: Definitions, comparisons, and research bases. *Journal of Engineering Education*, 95, 123–138.

Roselli, R. J., & Brophy, S. P. (2003). Redesigning a biomechanics course using challenge-based instruction. *IEEE Engineering in Medicine and Biology Magazine*, 22(4), 66–70.

Roselli, R. J., & Brophy, S. P. (2006). Experiences with formative assessment in engineering classrooms. *Journal of Engineering Education*, 95(4), 325–333.

Roselli, R. J., Howard, L., & Brophy, S. P. (2006). A computer-based free body diagram assistant. *Computer Applications in Engineering Education*, 14(4), 281–290.

Ruiz-Primo, M. A., Shavelson, R. J., Hamilton, L., & Klein, S. (2002). On the evaluation of systemic science education reform: Searching for instructional sensitivity. *Journal of Research in Science Teaching*, 39, 369–393.

Santiago-Román, A. I. (2009). *Fitting cognitive diagnostic assessment to the concept assessment tool for statics (CATS)* (Doctoral dissertation). Purdue University, Lafayette, IN.

Santiago-Román, A. I., Streveler, R. A., & DiBello, L. (2010). *The development of estimated cognitive attribute profiles for the Concept Assessment Tool for Statics*. Paper presented at the 40th ASEE/IEEE Frontiers in Education Conference, Washington, DC.

Santiago-Román, A. I., Streveler, R. A., Steif, P., & DiBello, L. V. (2010). *The development of a Q-matrix for the Concept Assessment Tool for Statics*. Paper presented at the ERM Division of the ASEE Annual Conference and Exposition, Louisville, KY.

Schum, D. (1987). *Evidence and inference for the intelligence analyst*. Lantham, MD: University Press of America.

Schwartz, D. L., & Arena, D. (2009). *Choice-based assessments for the digital age*. Stanford University, School of Education. Retrieved from http://courses.ischool.berkeley.edu/i290-pm4e/f10/sites/default/files/assessment_schwartz.pdf

Schwartz, D. L., Bransford, J. D., & Sears, D. (2005). Efficiency and innovation in transfer. In J. P. Mestre (Ed.), *Transfer of learning from a modern multidisciplinary perspective* (pp. 1–51). Greenwich, CT: Information Age.

Smith, K. A., Johnson, D. W., Johnson, R. W., & Sheppard, S. D. (2005). Pedagogies of engagement: Classroom-based practices. *Journal of Engineering Education*, 94(1), 87–101.

Steif, P. S., & Dantzler, J. A. (2004). A statics concept inventory: Development and psychometric analysis. *Journal of Engineering Education*, 94(4), 363–371.

Steif, P. S., & Hansen, M. A. (2006). Comparisons between performances in a statics concept inventory and course examinations, *International Journal of Engineering Education*, 22, 1070–1076.

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036

Steif, P. S., & Hansen, M. A. (2007). New practices for administering and analyzing the results of concept inventories. *Journal of Engineering Education*, *96*, 205–212.

Stigler, S. M. (1982). Thomas Bayes' Bayesian inference. *Journal of the Royal Statistical Society*, *Series* A, *145*, 250–258.

Toulmin, S. E. (2003). *The uses of argument*. Cambridge: Cambridge University Press.

van der Linden, W. J., & Hambleton, R. K. (Eds.) (1996). *Handbook of modern item response theory*. New York, NY: Springer.

Walker, J. T., Cordray, D. S., King, P. H., & Brophy, S. P. (2006). Design scenarios as an assessment of adaptive expertise. *International Journal of Engineering Education*, *22*(1), 1–7.

Wiggins, G. (1998). *Educative assessment: Designing assessments to inform and improve student performance*. San Francisco, CA: Jossey-Bass.

Woods, D. R. (1997). Developing problem-solving skills: The McMaster problem solving program. *Journal of Engineering Education*, *86*(2), 75–92.

Downloaded from https://www.cambridge.org/core. University of Warwick, on 21 Oct 2018 at 04:29:36, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/CBO9781139013451.036